

Case 30: Multiple-ignition topology

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE30.MIT. This case verifies that the standard ELMFIRE executable initializes, schedules, transports, and merges multiple point ignitions without depending on source-tree test drivers. It preserves the scientific intent of Adam Laird’s dual-ignition, staggered spot-merging, and near-boundary component scenarios.

2 Mathematical formulation and numerical implementation

Each ignition introduces a local negative level-set region at position (x_i, y_i) and time t_i . ELMFIRE then advances ϕ according to $\partial_t \phi + V_n |\nabla \phi| = 0$. With homogeneous fuel, zero slope, and zero wind, simultaneous mirrored sources must remain reflection symmetric. Adding uniform wind must produce a measurably different arrival-time solution. Delayed sources must appear at their requested times, and clipping at a domain boundary must not suppress or wrap a source.

3 Derivation of expected behavior

The no-wind forcing and ignition pair are invariant under reflection about their mapped raster axes; therefore the discrete arrival-time solution has the same symmetries. Because ELMFIRE maps coordinates to one-based cells with a ceiling operation, the evaluator reflects the largest balanced row window about the actual ignition row rather than an intercell domain line. The wind variant changes only the wind rasters, so a nonzero solution difference is a metamorphic check of weather coupling. For scheduled ignition i , the containing raster cell should receive an arrival time within one 1200 s of t_i . The eight boundary sources are placed one five-cell margin inside the square domain and all must initialize.

For any two arrival-time arrays A and B , the implemented comparison is

$$E(A, B) = \frac{\text{mean}_{i \in V} |A_i - B_i|}{\max[P_{90}(A_i : i \in V), 1 \text{ s}]}, \quad V = \{i : A_i, B_i \text{ are finite}\}.$$

For the two simultaneous ignitions at $(-52.5, 0)$ and $(52.5, 0)$ m, the reflection reference is exact symmetry, $E = 0$; the metric is the larger of the left–right and centered top–bottom reflection errors and must satisfy $E \leq 0.02$. The 5 mph, 225-degree wind run is compared with the no-wind run by the same formula and must give $E \geq 0.05$, demonstrating a resolved forcing response. For scheduled ignitions (x, y, t) equal to $(0, 600, 0)$, $(0, 300, 1200)$, $(0, 0, 2400)$, $(0, -300, 3600)$, and $(0, -600, 4800)$, the timing metric is

$$E_t = \max_j |TOA(x_j, y_j) - t_j| \leq 1200 \text{ s}.$$

Finally, the count metric is $\sum_{j=1}^8 \mathbf{1}[TOA(x_j, y_j) \leq 1200 \text{ s}] = 8$ at the eight generated edge/corner points $(\pm 975, 0)$, $(0, \pm 975)$, and $(\pm 975, \pm 975)$ m.

4 Verification metrics and acceptance criteria

Reflection error is the larger normalized mean absolute difference between the no-wind arrival field and its reflection about each mapped ignition axis; it must not exceed 0.02. Wind response is the normalized mean absolute difference between the wind and no-wind fields and must be at least 0.05. The maximum scheduled-source timing error must not exceed one output interval. All eight boundary-source cells must have finite arrival times no later than one output interval. These metrics were selected because they test invariants of the coupled ignition/level-set implementation without reproducing ELMFIRE’s internal algorithms.

5 Simulation configuration and predicted outcome

All variants use a 5 m square grid in EPSG:32610, Fuel Model 10, 3/5/6 percent dead-fuel moisture, and flat terrain. The variants contain two simultaneous no-wind sources, the same pair under 5 mph wind from 225 degrees, five staggered sources, and eight edge/corner sources. In the staggered variant the first source is anchored at ($t=0$) and the other four follow at 1200-s intervals. This common translation of the original absolute schedule preserves every relative delay while preventing versions that have no active front from advancing past a strictly future first ignition. The predicted fields are respectively symmetric merger, wind-distorted merger, correctly timed merger, and eight clipped but valid boundary fires.

Parameter	Configured value	Units	Role
Domain/grid	2000 m square; 5	m/cell	Resolves source placement and merging
Fuel/moisture	FM10; 3/5/6	— / percent	Homogeneous surface response
Time control	step 1, maximum 5; stop 14400	s	Exact timestep multiple for scheduled sources
Variants	dual calm, dual wind, staggered, boundary	—	Exercises symmetry, weather, timing, and clipping
Output cadence	1200	s	Timing-error acceptance scale

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

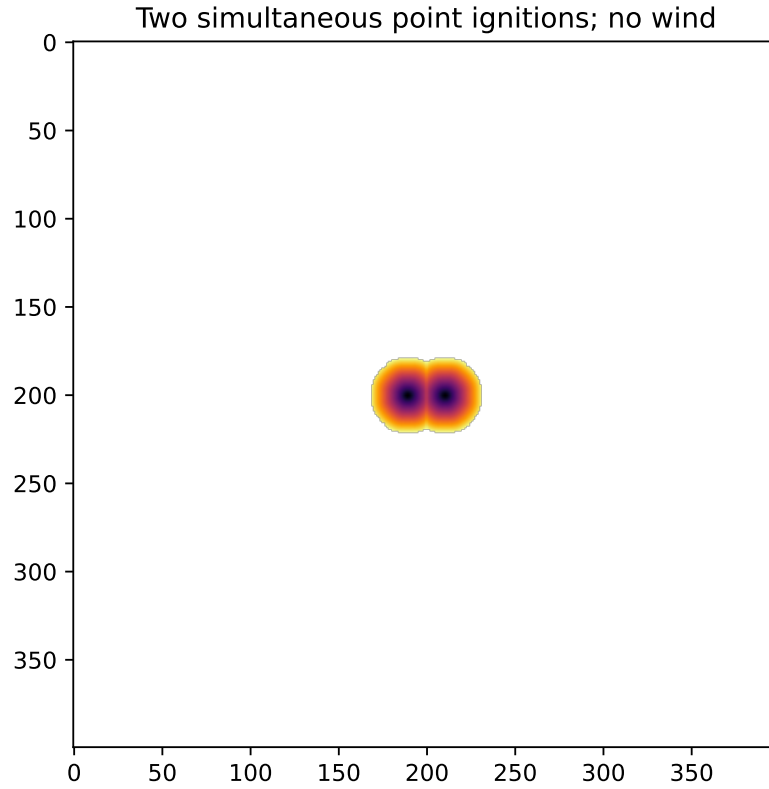


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated result is **PASS**; 4 of 4 required metrics pass across 4 variants. Missing ELMFIRE output yields NOT EVALUABLE rather than a pass.

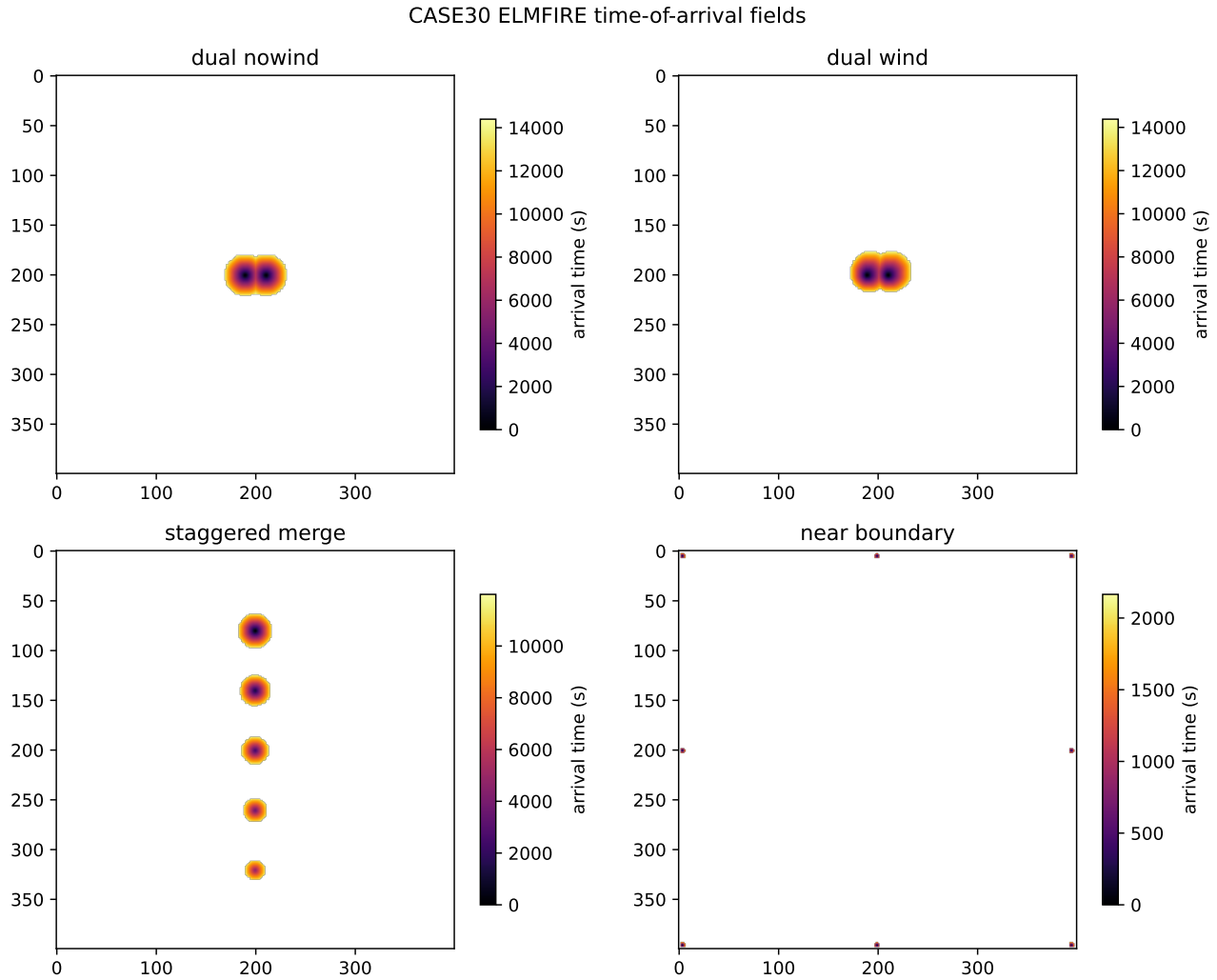


Figure 2: Case-local time-of-arrival fields for the four ignition-topology variants.

7 Calculated metrics and verification decision

Metric	Expected	Calculated	Status
dual-ignition reflection error	≤ 0.02 (2% normalized L1)	0.000472284	PASS
wind-forcing response	≥ 0.05 (5% normalized L1)	0.0937636	PASS
maximum scheduled-ignition timing error	≤ 1200 (one output interval)	0 s	PASS
boundary ignitions initialized	8 (all eight points)	8 count	PASS

The authoritative decision is **PASS** from `outputs/metrics.json`.

8 Technical interpretation

A symmetry failure points to directional indexing or level-set transport. No wind response indicates that the weather field did not reach fire behavior. Timing error implicates the scheduled-ignition path. Missing edge ignitions indicates boundary-index clipping or coordinate conversion failure.

9 Scope and limitations

This is deterministic implementation verification, not fire-behavior validation. It does not expose the internal Fortran ignition routines directly; it checks their observable behavior through an unmodified ELMFIRE executable.